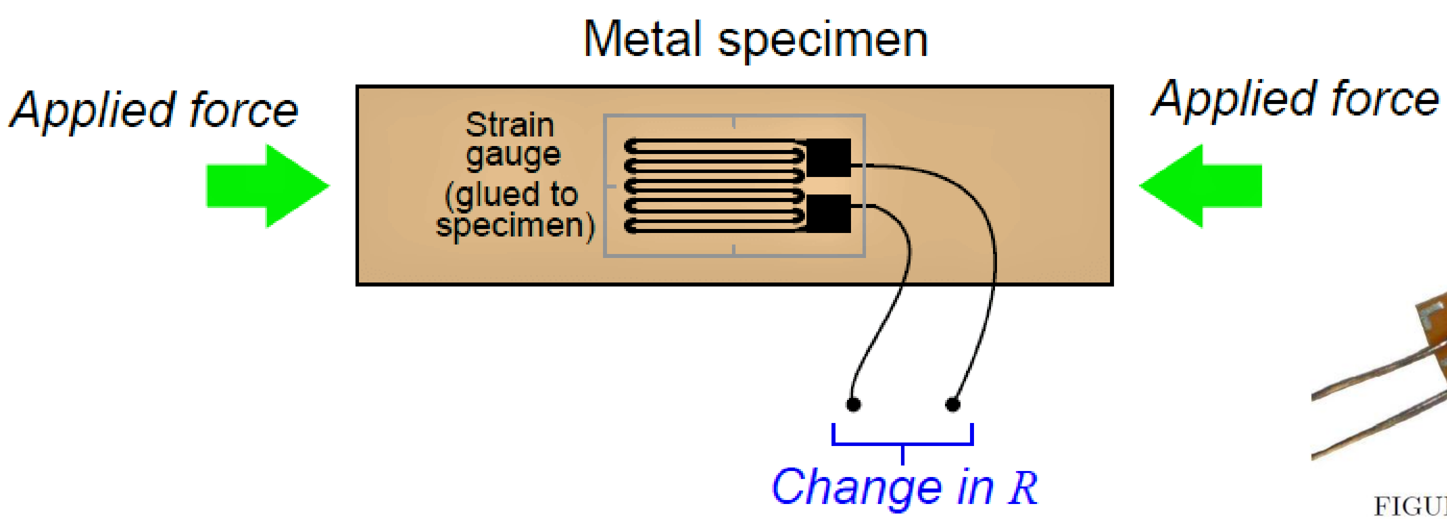
Force

Metallic strain gauges

- The electric resistance of a wire

$$R = \rho \frac{l}{A}$$

Material with a very small temperature coefficient of electrical resistance.



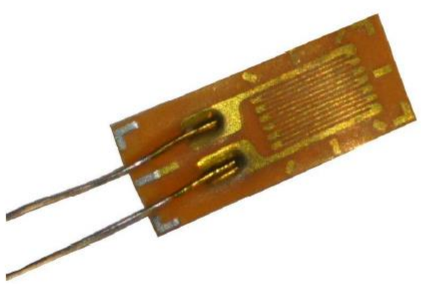
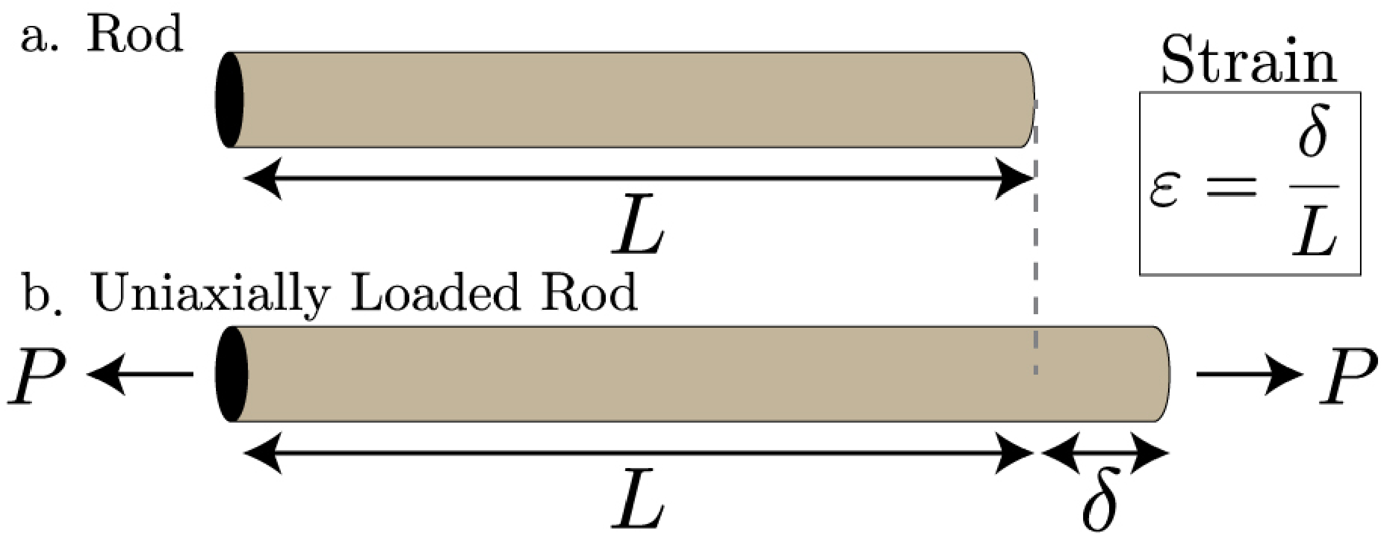


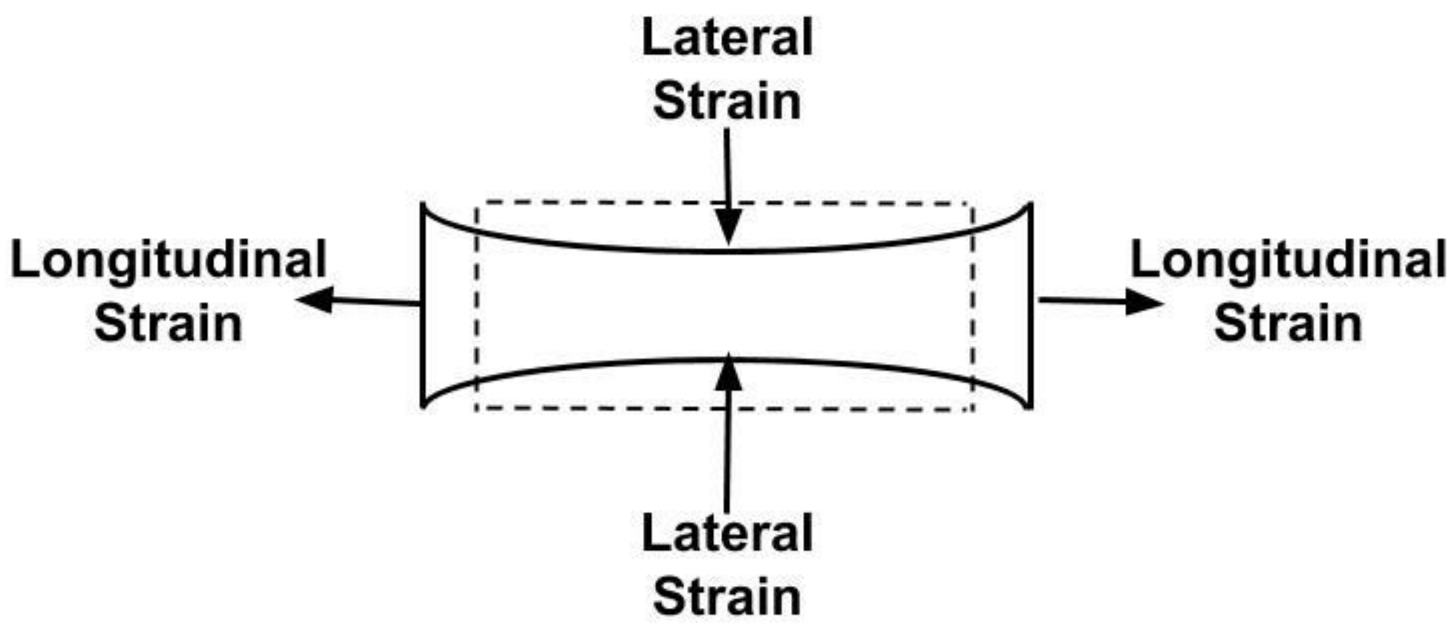
FIGURE 5.1: Foil strain gauge—photo author

Strain

Strain (m/m): the measure of how much an object is stretched or deformed

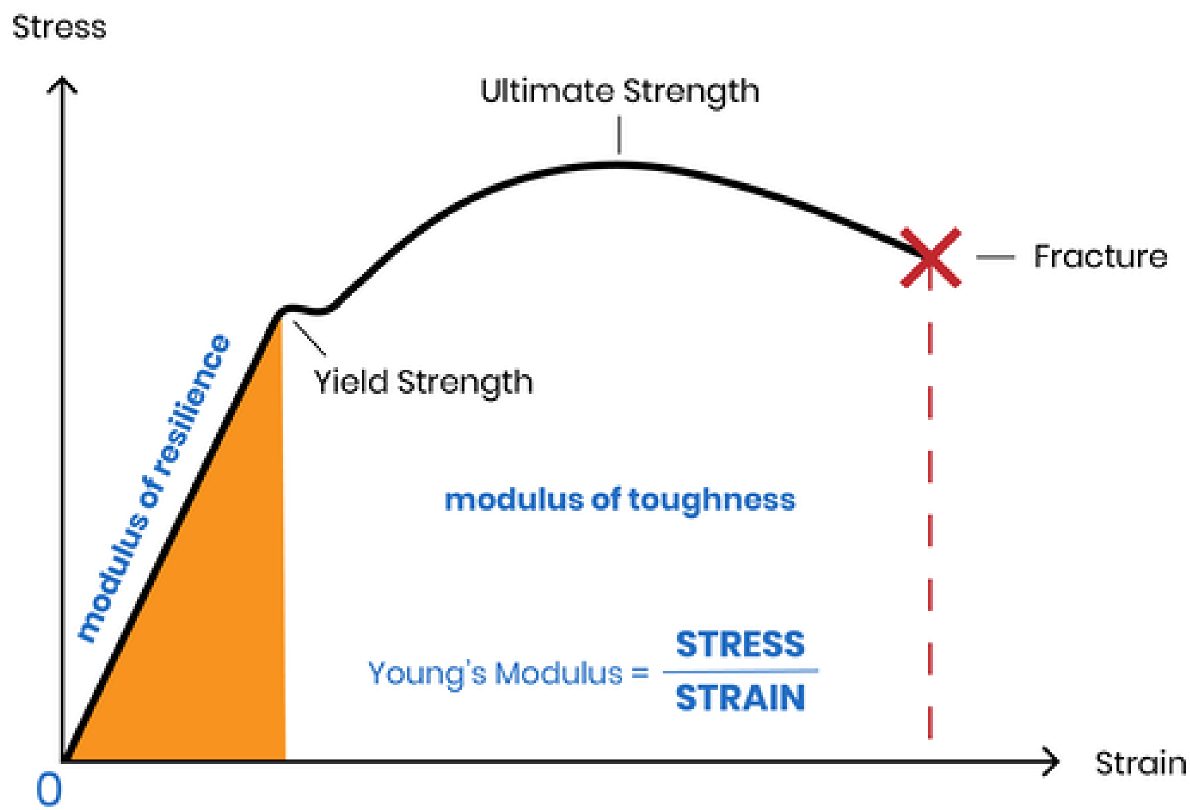


Poisson's Ratio

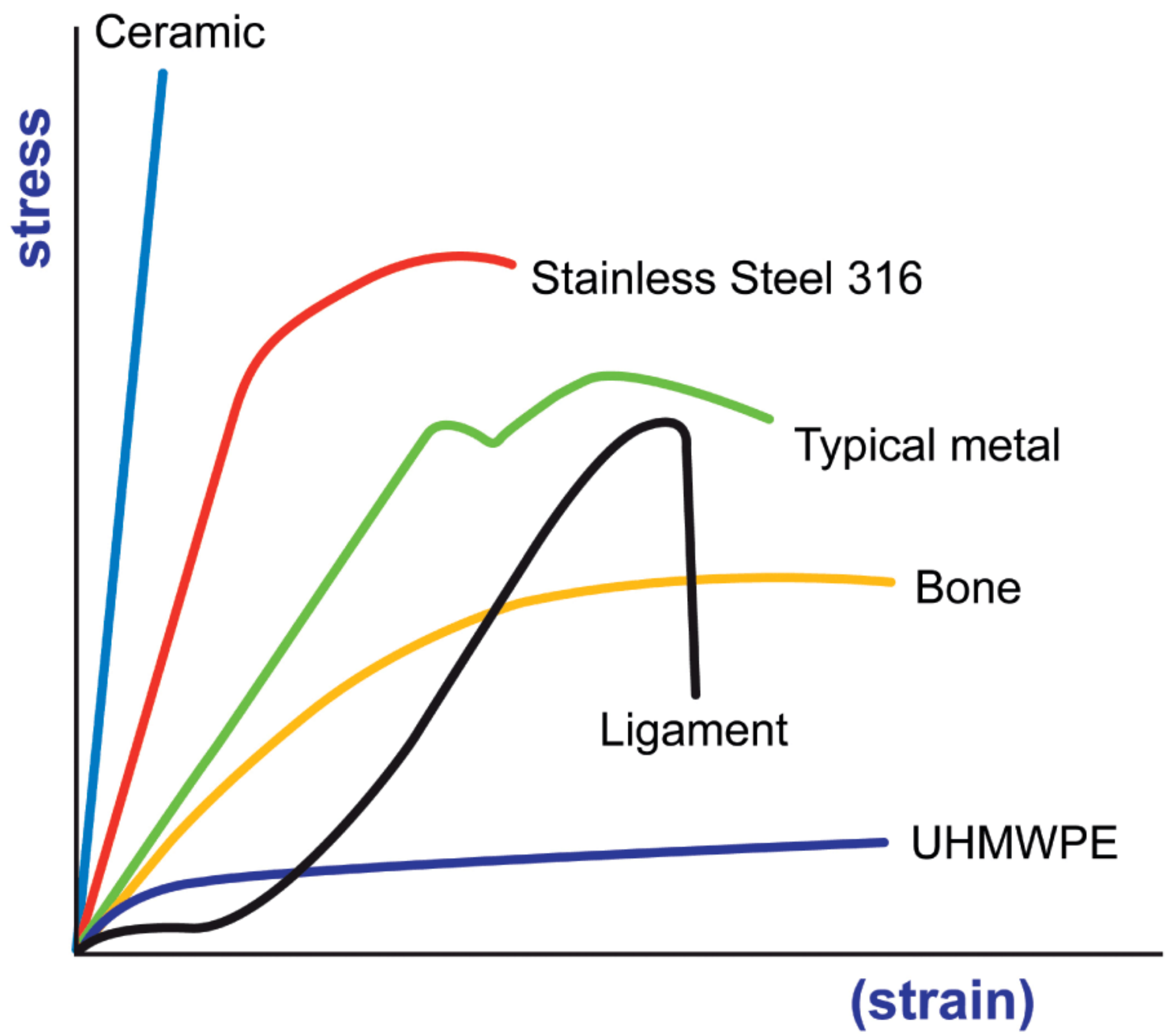


$$\text{Poisson's Ratio} = \frac{\text{Lateral Strain}}{\text{Longitudinal Strain}}$$

Young's Modulus



Young’s Modulus



Metallic strain gauges

- Change of electrical resistance depends on:
  - Change of electrical resistance with strain
  - Change of electrical resistance with change of length
  - Change of electrical resistance with change of cross

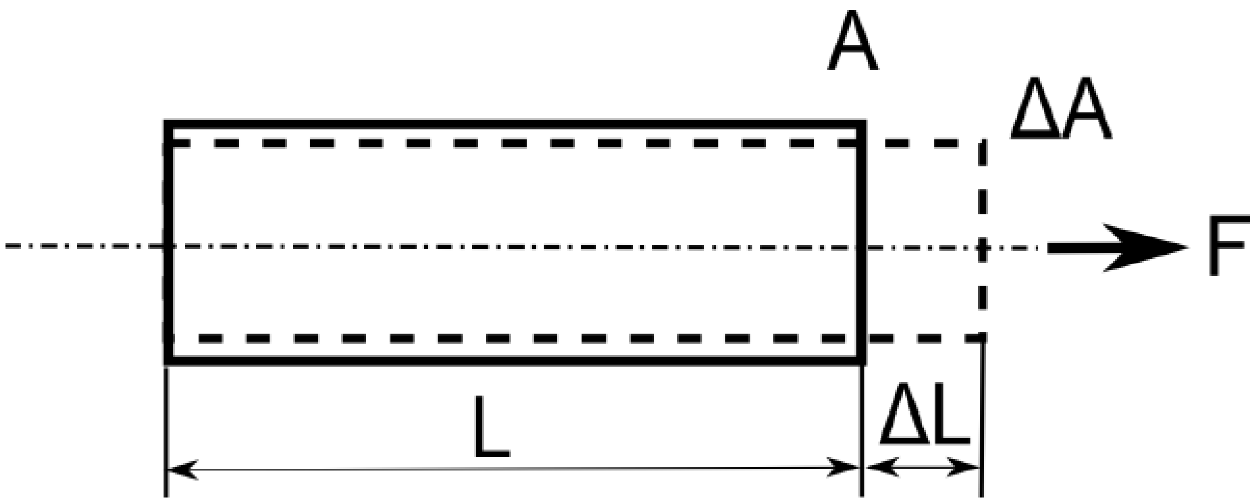


FIGURE 5.3: Change of dimensions with applied force

Metallic strain gauges

$$\frac{dR}{R} = \frac{dl}{l} - \frac{dA}{A} + \frac{d\rho}{\rho}$$

Dependence of length and cross section (Poisson ratio ν)

$$\frac{dA}{A} = -2\nu \frac{dl}{l}$$

$$\frac{dR}{R} = \frac{dl}{l} - \frac{dA}{A} + \frac{d\rho}{\rho} = (1 + 2\nu)\frac{dl}{l} + \frac{d\rho}{\rho}$$

$$\frac{dR}{R} = \underbrace{\left[(1 + 2\nu) + \frac{\frac{d\rho}{\rho}}{\frac{dl}{l}}\right]}_{\text{Gauge factor}} \cdot \frac{dl}{l} = K \cdot \varepsilon$$

Metallic strain gauges

- **Gauge factor (sensitivity of the gauge)**

The ratio of per unit change in resistance to per unit change in length.

$$G_f = \frac{\Delta R/R}{\Delta L/L} = 1 + 2\nu + \frac{\Delta\rho/\rho}{\varepsilon}$$

Metallic strain gauges

TABLE 5.1: Gauge factor for selected materials—based on data from [35]

Material	Gauge factor
Pt 100%	6.1
Pt 95%, Ir 5%	5.1
Pt 92%, W 8%	4.0
Isoelastic (Fe 55.5%, Ni 36% Cr 8%, Mn 0.5%)	3.6
Constantan / Advance / Copel (Ni 45%, Cu 55%)	2.1
Nichrome V (Ni 80%, Cr 20%)	2.1
Karma (Ni 74%, Cr 20%, Al 3%, Fe 3%)	2.0
Armour D (Fe 70%, Cr 20%, Al 10%)	2.0
Monel (Ni 67%, Cu 33%)	1.9
Manganin (Cu 84%, Mn 12%, Ni 4%)	0.47
Nickel (Ni 100%)	−12.1

Strain-gauge rosettes

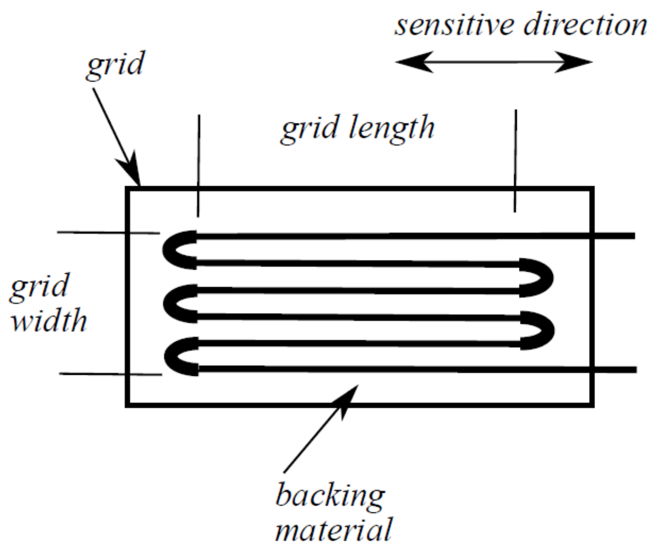
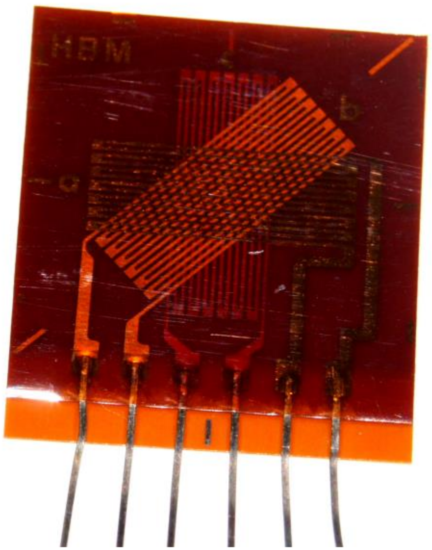
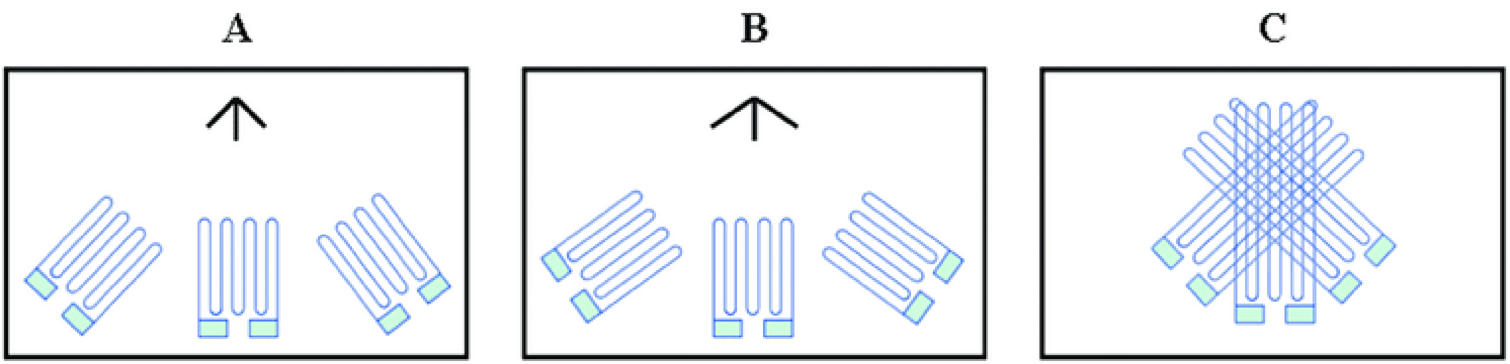


FIGURE 5.6: Main components of a foil strain gauge





Temperature compensation

- A steel bar with dimensions 10 × 10 mm, Young's modulus E = 200 GPa. The bar is loaded with a force F = 1000 N.

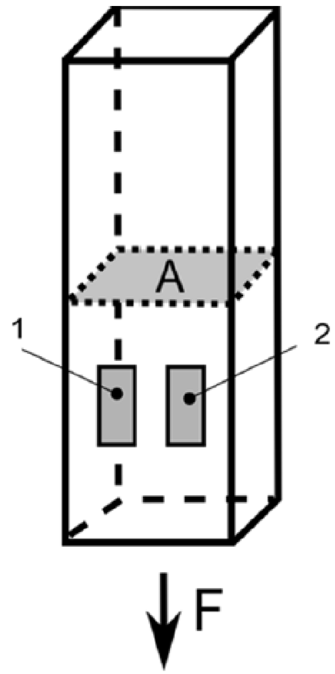


FIGURE 5.4: Experiment temperature compensation

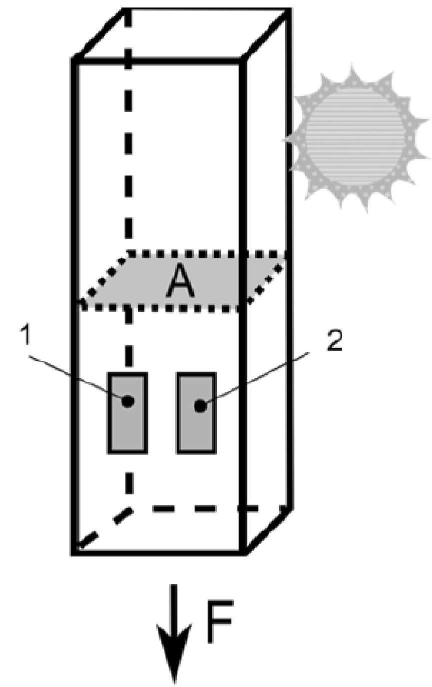


FIGURE 5.5: Experiment temperature compensation—heated bar

Temperature compensation

- For a strain gauge with R = 120, the absolute change of electrical resistance:

$$dR = K\epsilon R = 2 \times 50 \times 10^{-6} \times 120 = 12\ m\Omega$$

- The change of rresistance caused by the temperature change for the **constantan strain gauge**:

$$R(T) = R(T_0)(1 + \alpha\Delta T) = 120(1 + 8 \times 10^{-6} \times 10) = 120.0096\ \Omega$$

$$\Delta R(T) = 9.6\ m\Omega$$

- The change of rresistance caused by the temperature change for the **copper strain gauge**:

$$R(T) = R(T_0)(1 + \alpha\Delta T) = 120(1 + 3.9 \times 10^{-3} \times 10) = 124.68\ \Omega$$

$$\Delta R(T) = 4.68\ \Omega$$

Circuits for strain gauges

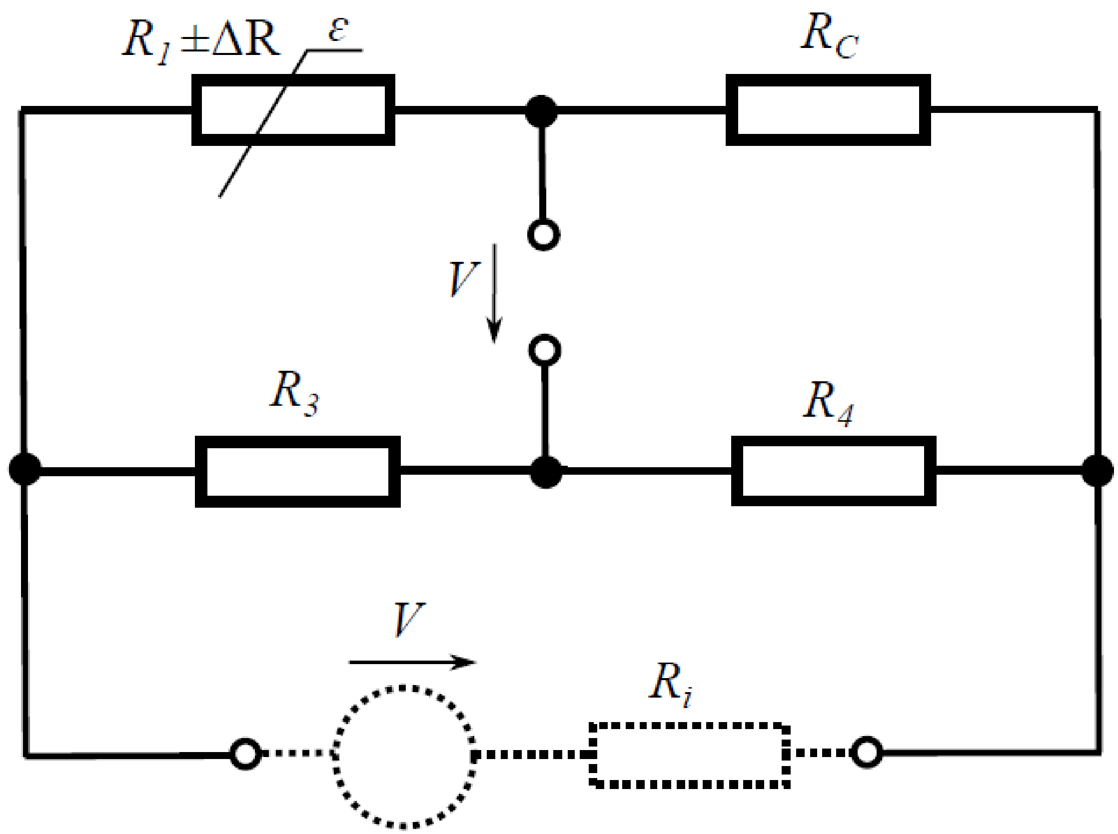


FIGURE 5.9: Strain gauge—quarter-bridge, R_1 = sensor, R_C = temperature compensation

Circuits for strain gauges

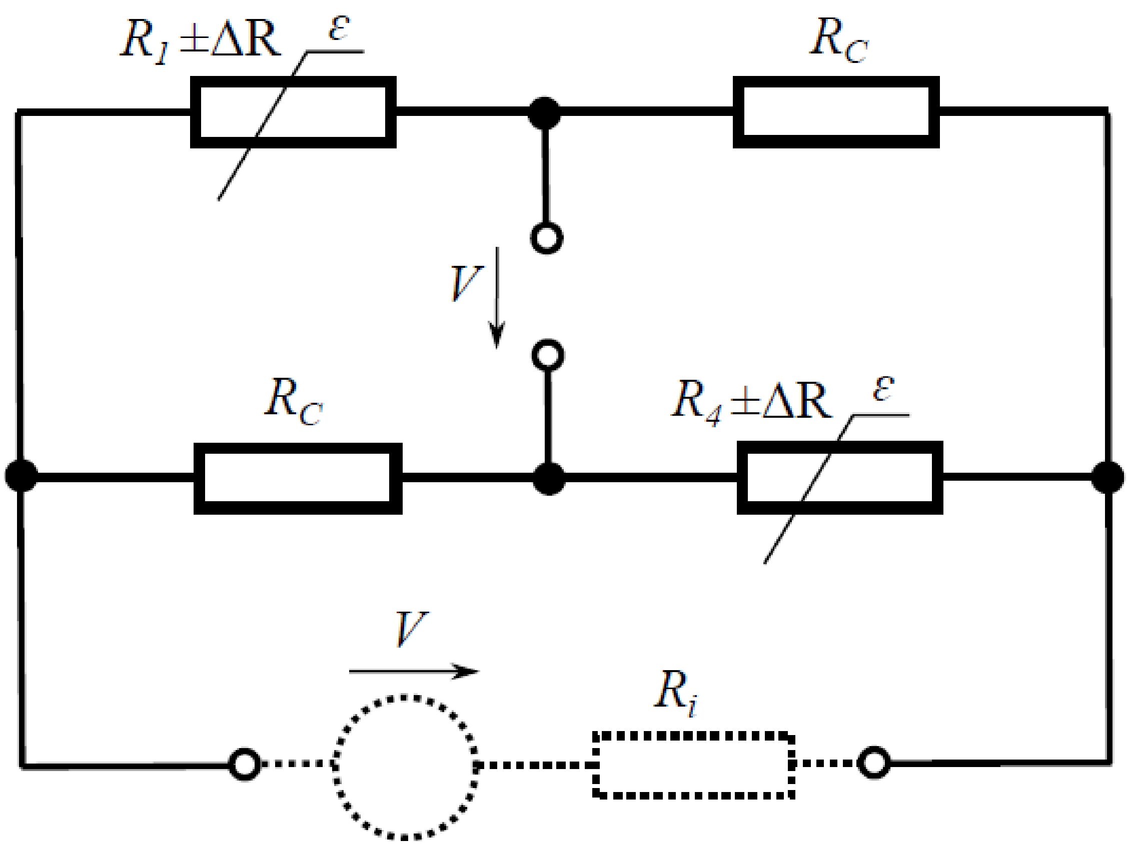


FIGURE 5.11: Half-bridge—sensors measure in the same direction, R_1, R_4 = sensor, R_C = temperature compensation

Circuits for strain gauges

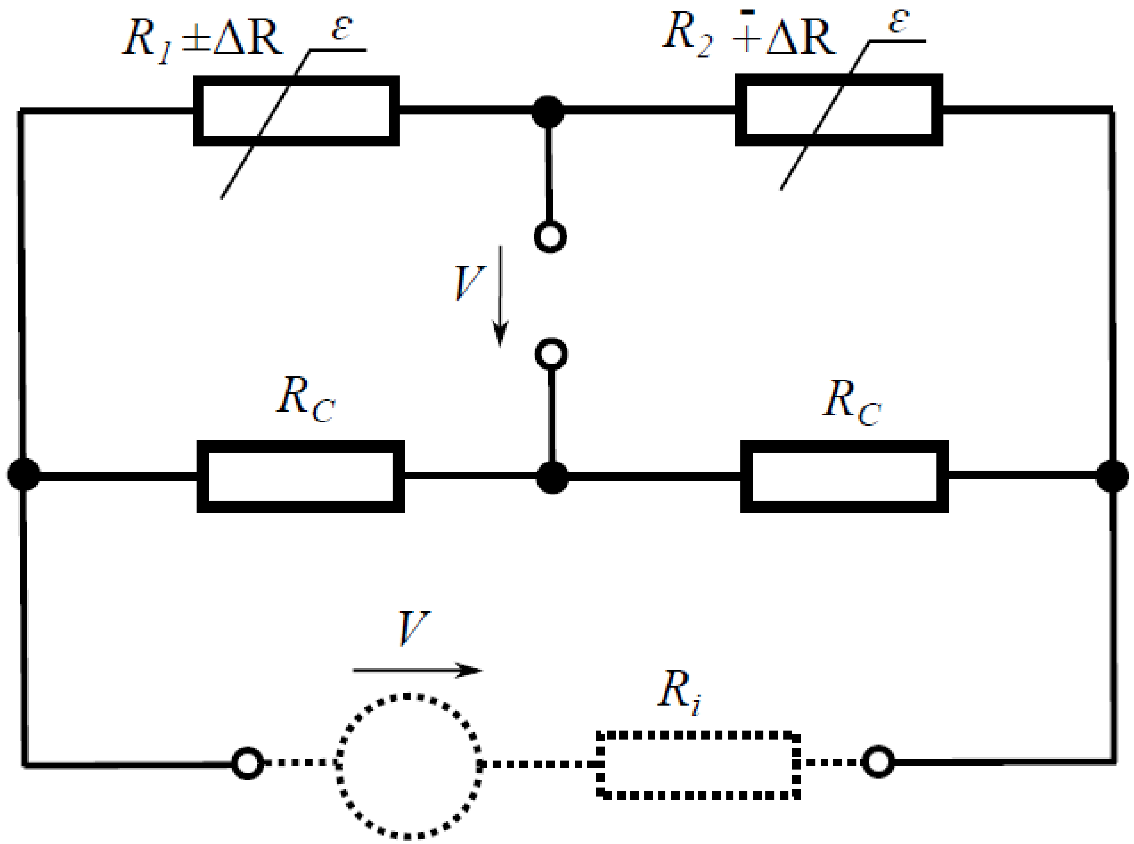


FIGURE 5.12: Half-bridge—sensors measure in the opposite direction, R_1, R_4 = sensor, R_C = temperature compensation

Circuits for strain gauges

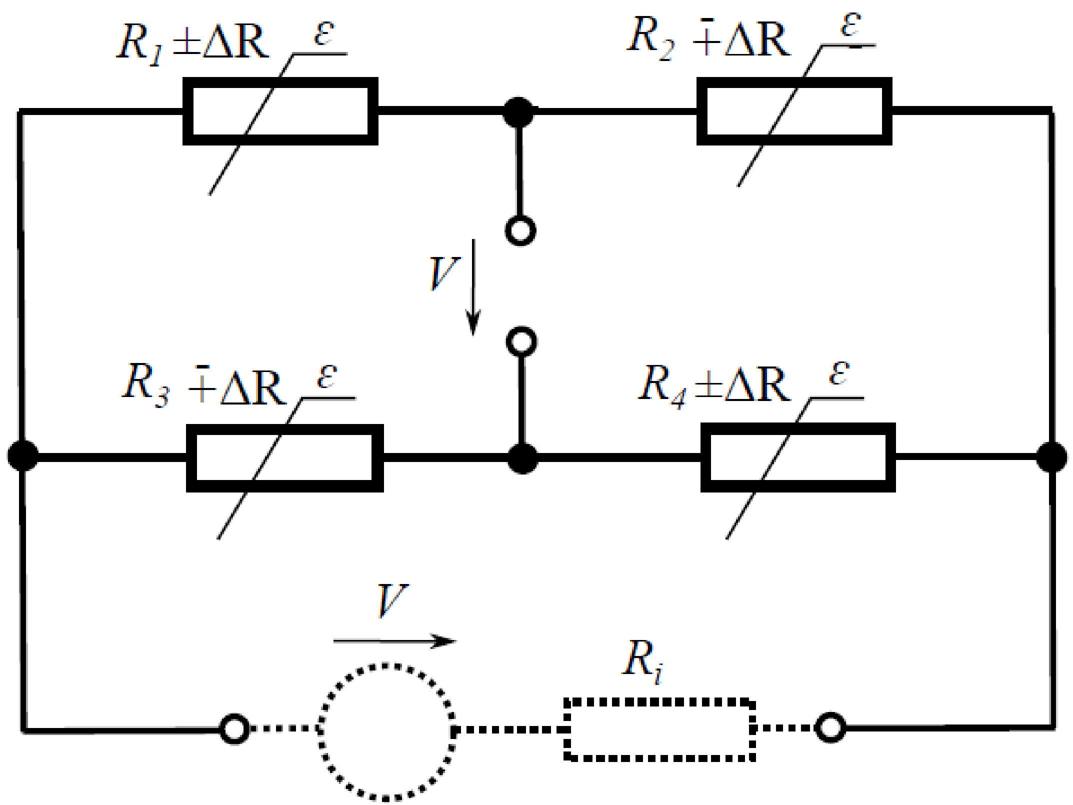
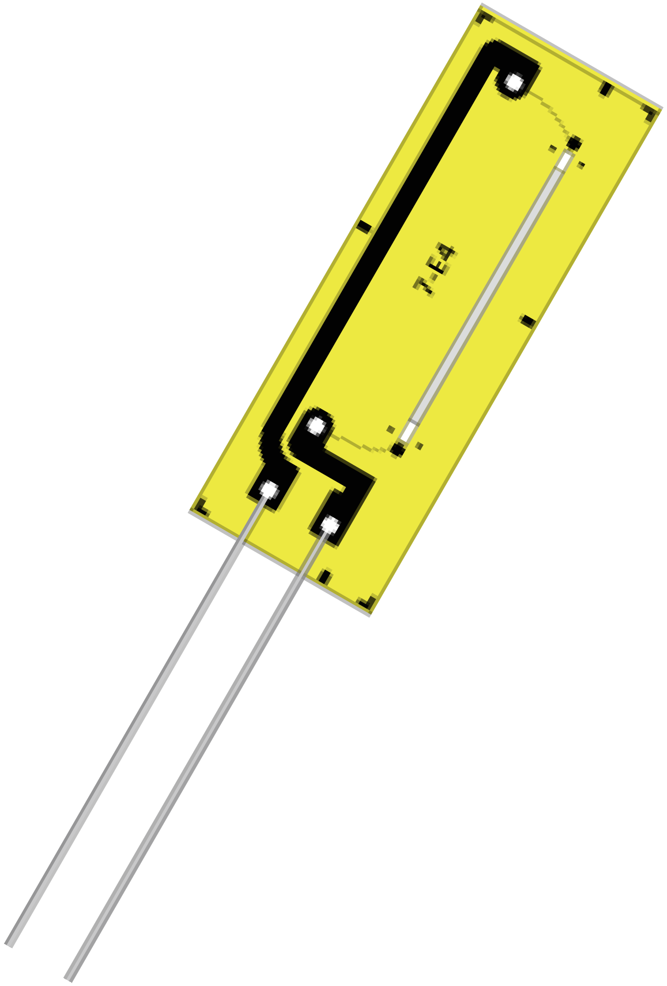
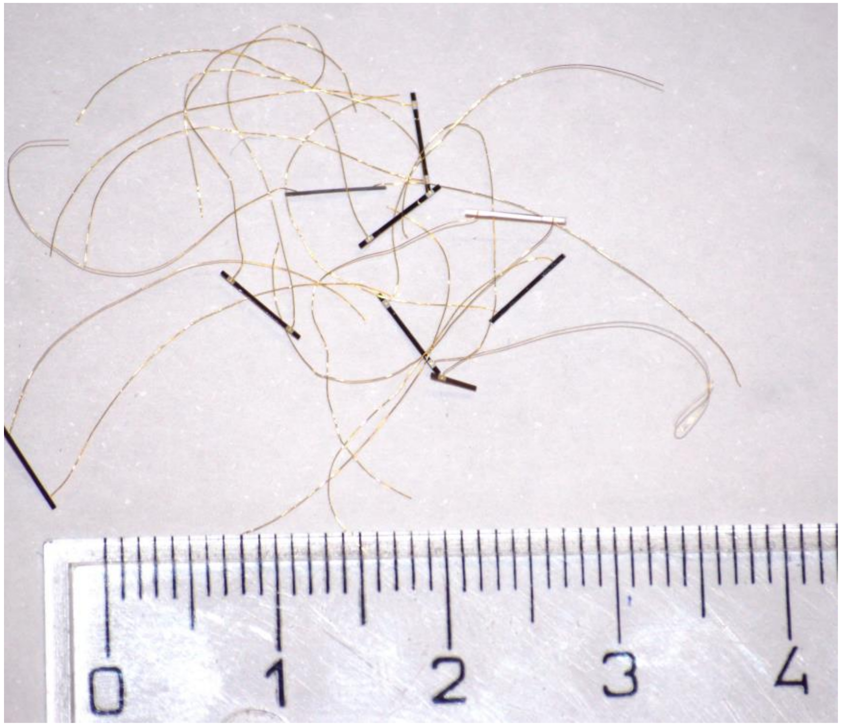


FIGURE 5.13: Full bridge

The full bridge doubles the sensitivity as compared to a half-bridge.

Semiconductor strain gauges





Semiconductor strain gauges

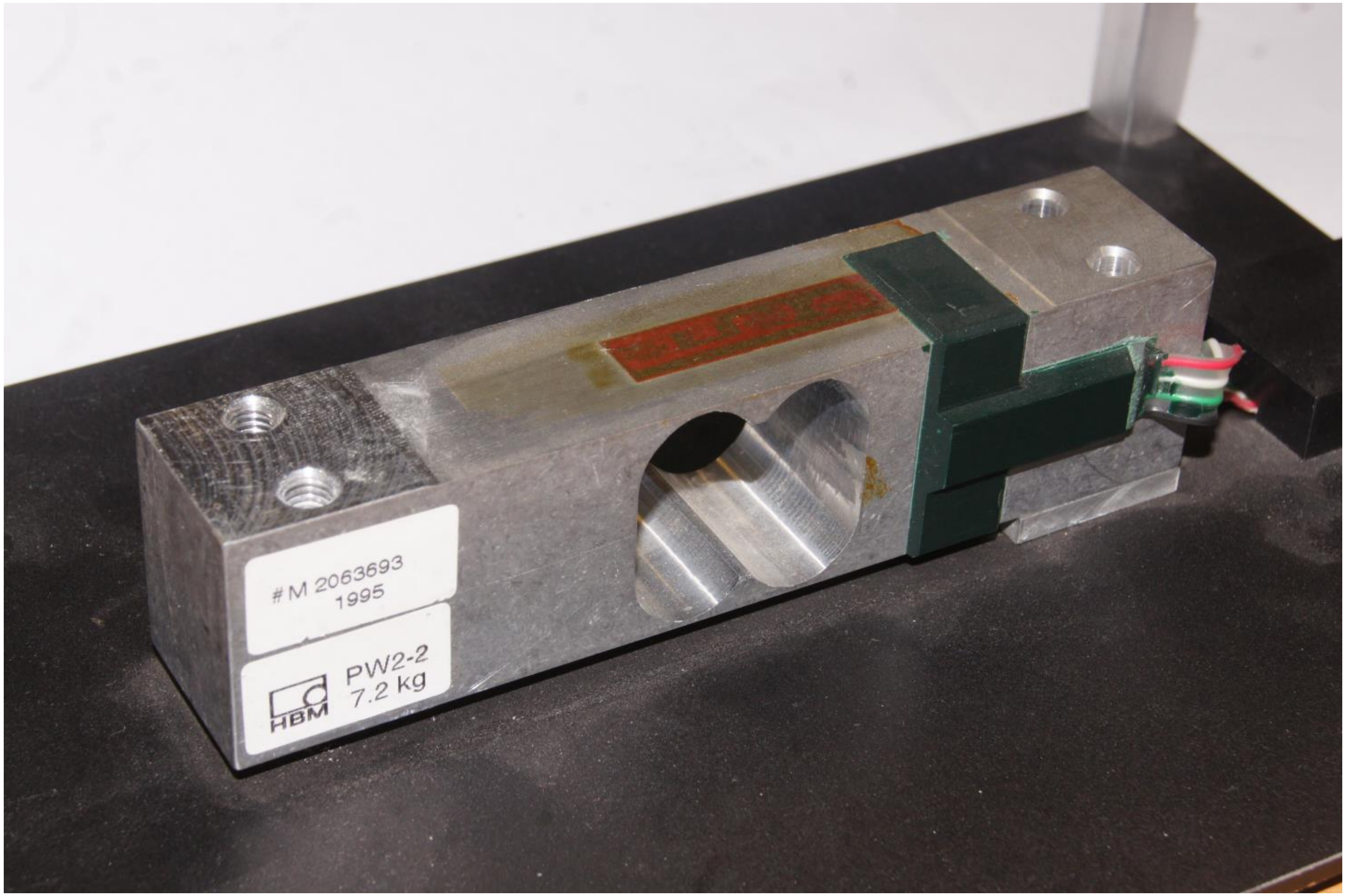
- Have significantly larger sensitivity (x25 larger than that of metal strain gauges.
- GF for a p-type semiconductor strain gauge is approximately +110 to +130; for n-type semiconductor strain gauges, GF is approximately -80 to -100

$$R_{0,t} = R_{0.25}(1 + a(t - 25) + b(t - 25)^2)$$

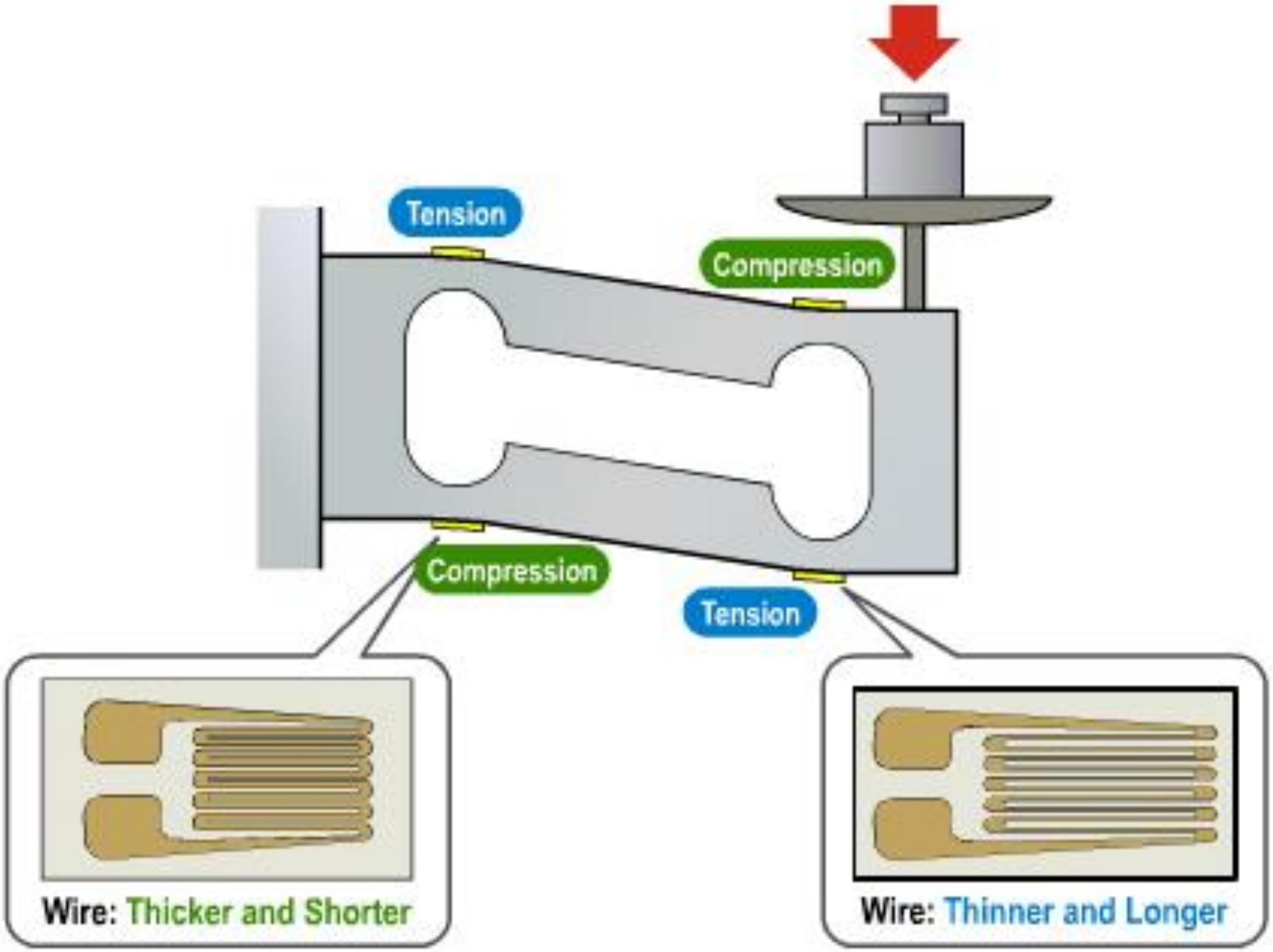
Semiconductor strain gauges are about 75x more sensitive to temperature.

$$R_{\epsilon,25} = R_{0.25}(1 + C_1\epsilon + C_2\epsilon^2)$$

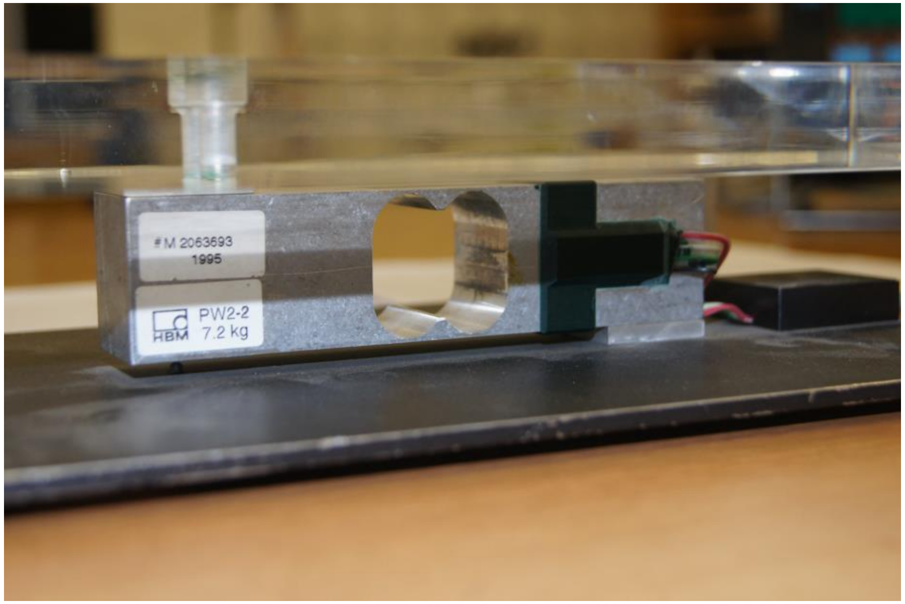
Load Cells



Load Cells


Tension
Compression
Compression
Tension
Wire: Thicker and Shorter
Wire: Thinner and Longer

Load Cells



Load Cells

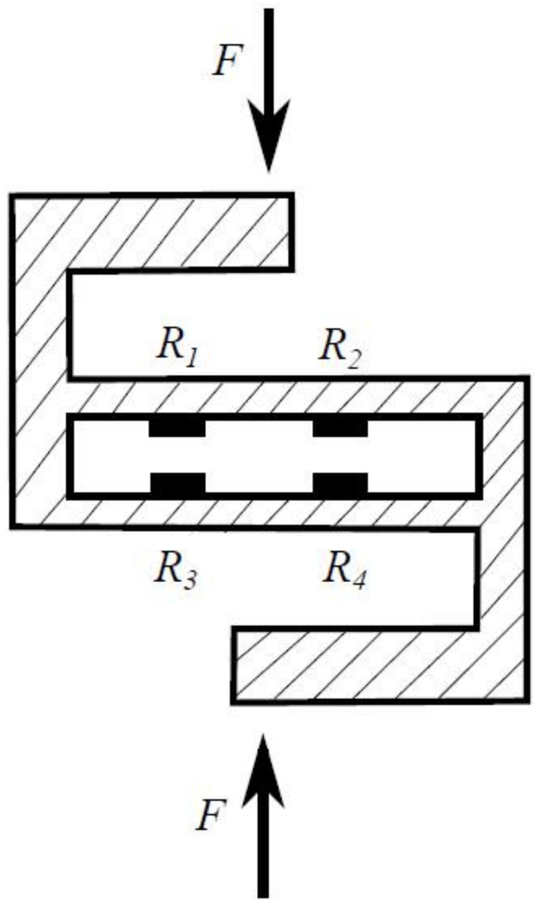
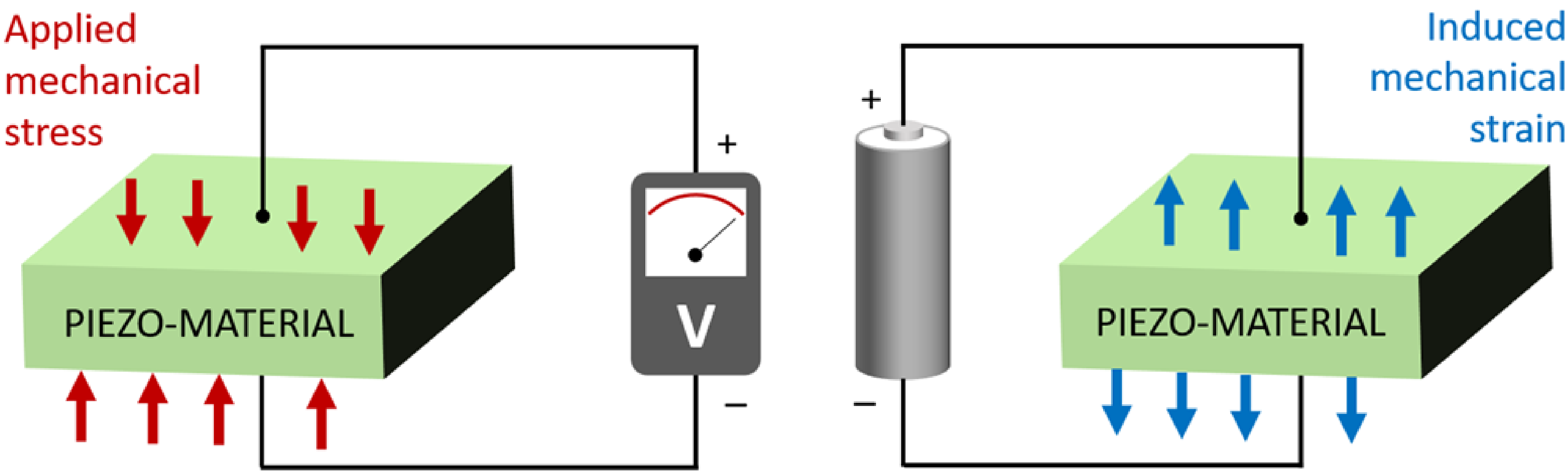


FIGURE 5.22: Type-S load cell

FIGURE 5.23: Type-S load cell—photo author

Piezoelectricty



Piezoelectricty

Quartz

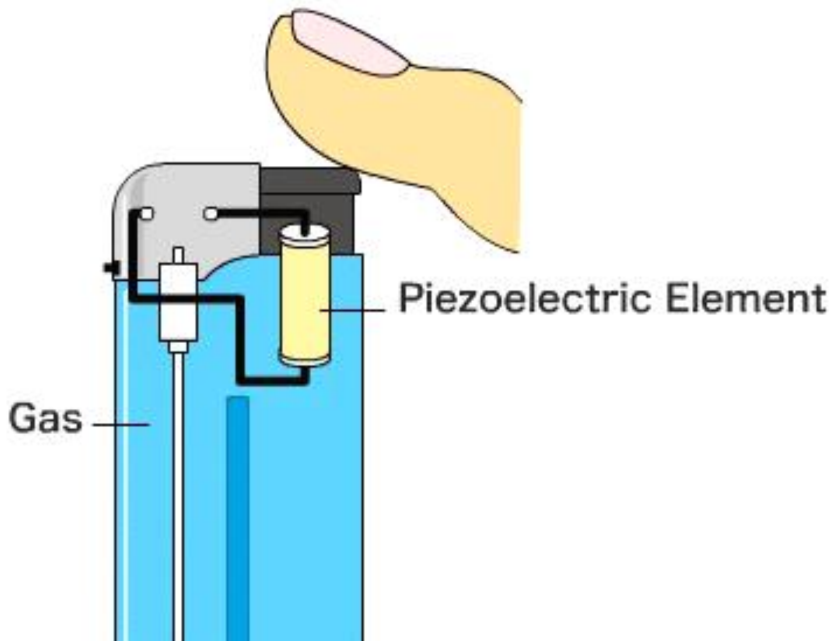
Barium titanate

Rochelle salt

Lead zirconium titanate (PZT)

Cane sugar

Piezoelectricty



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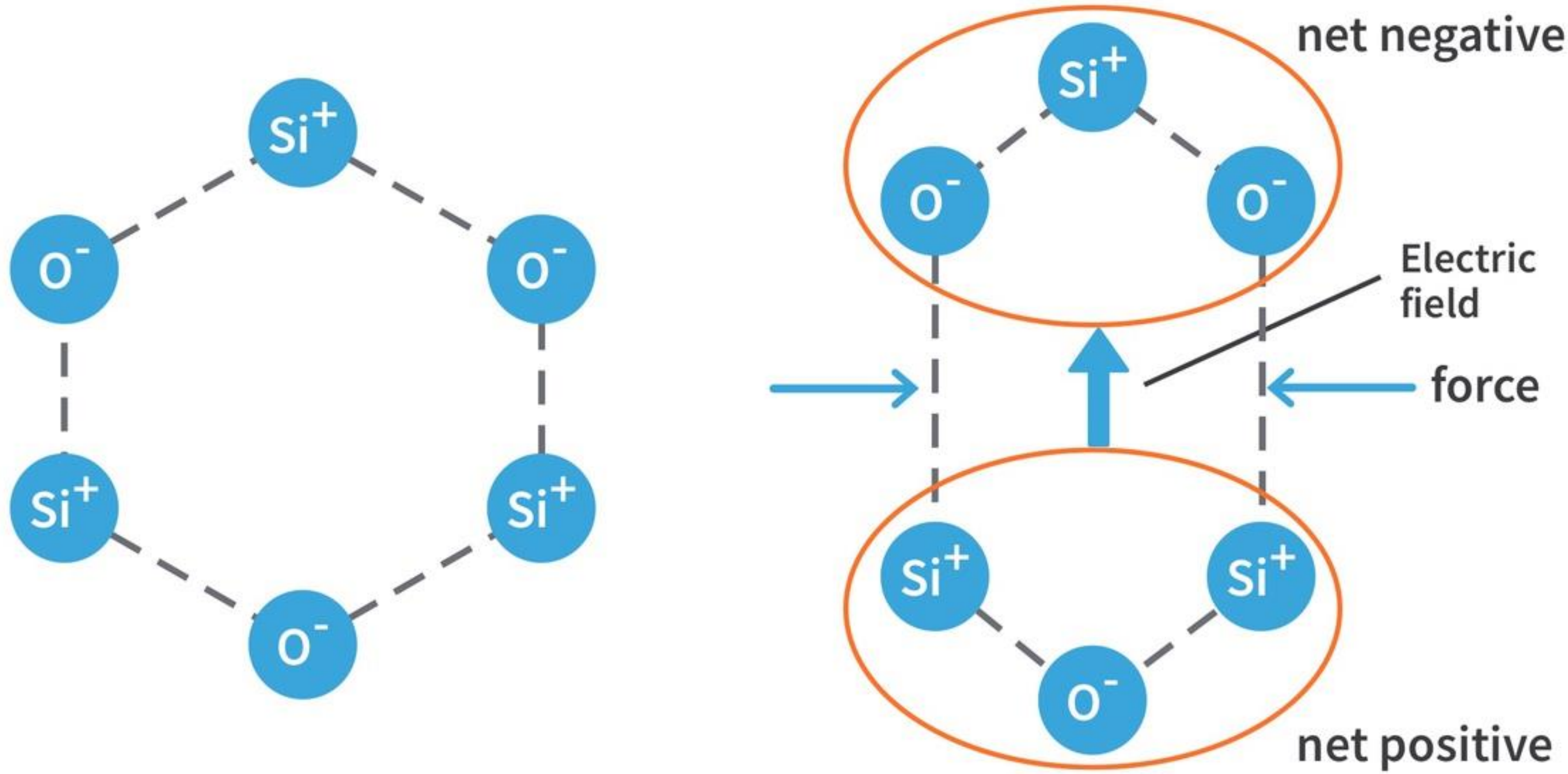

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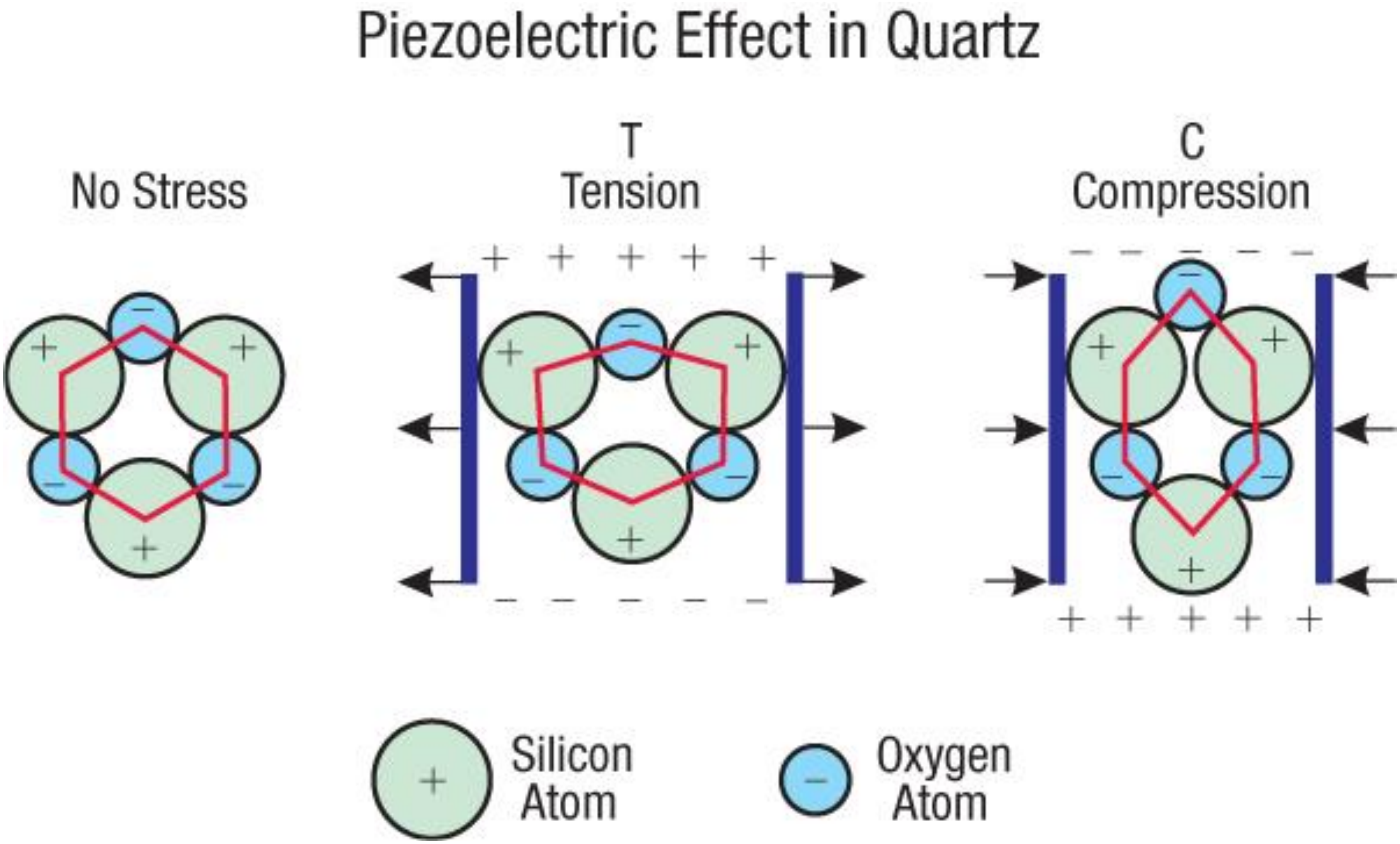

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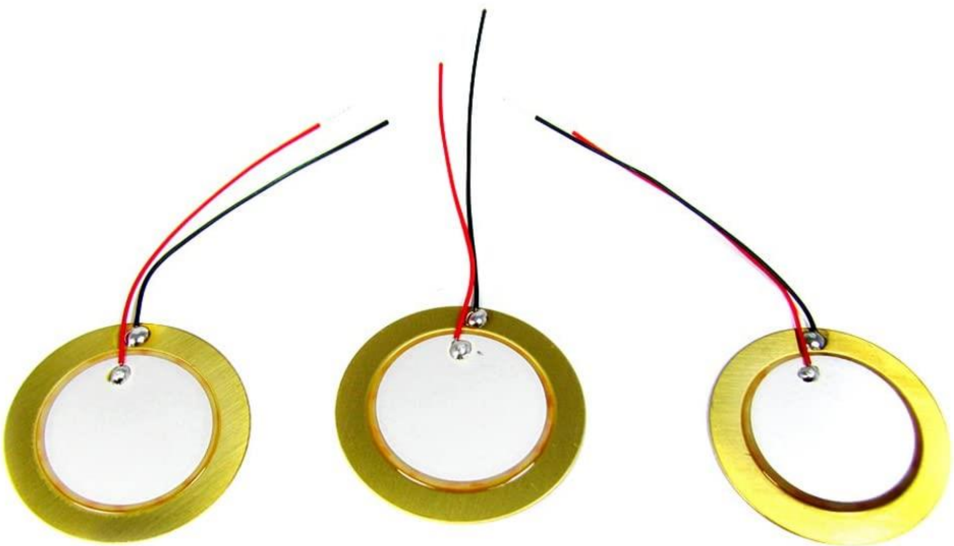
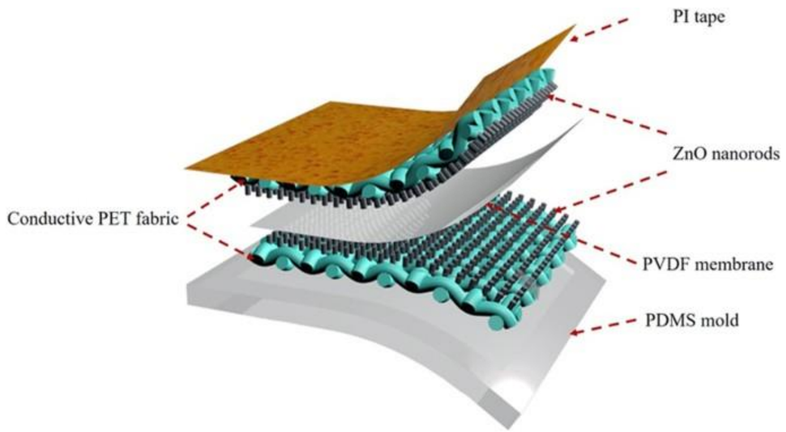
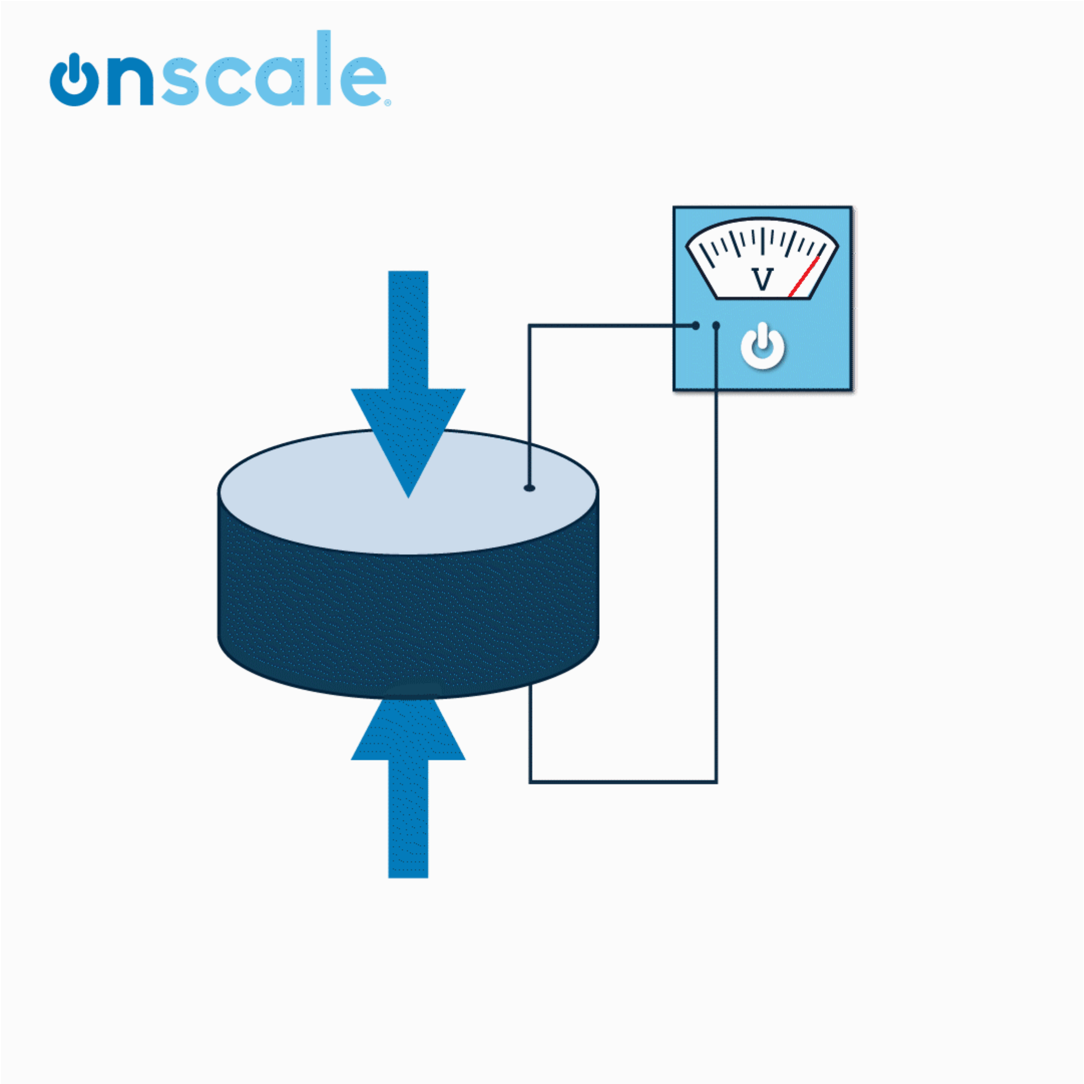




Piezoelectricty



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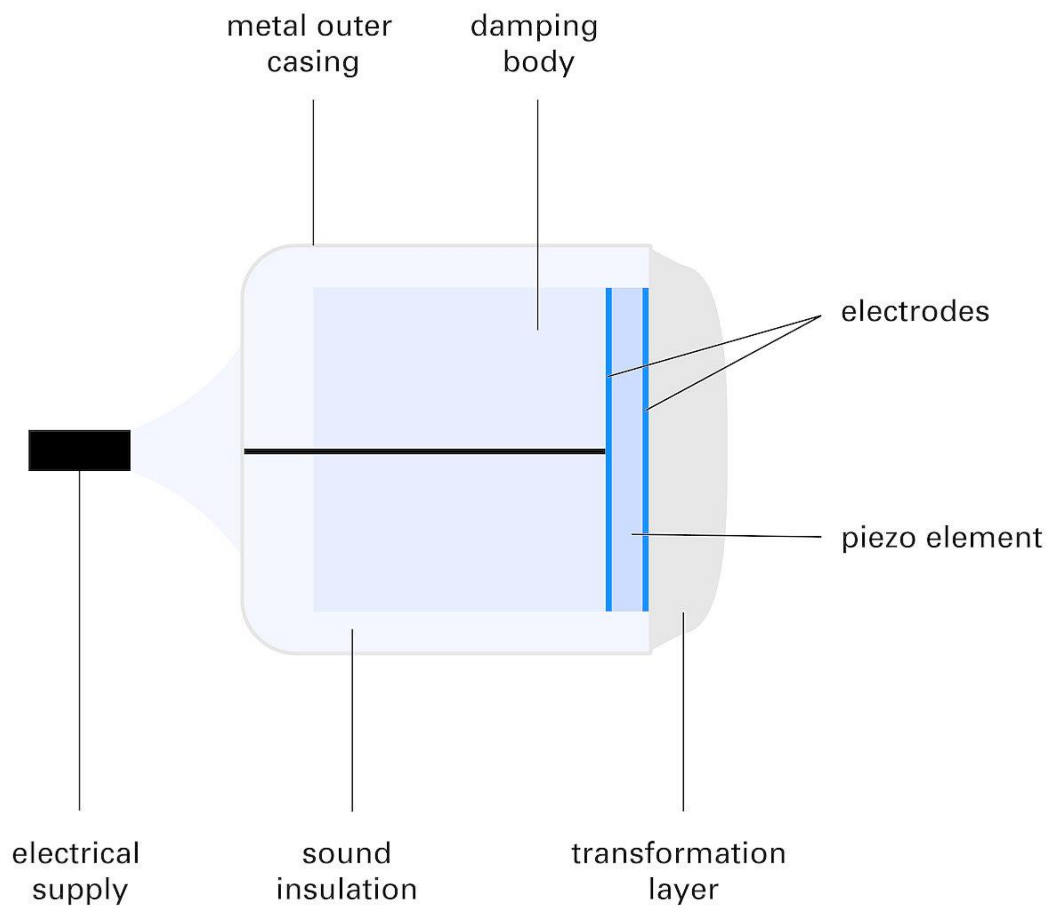

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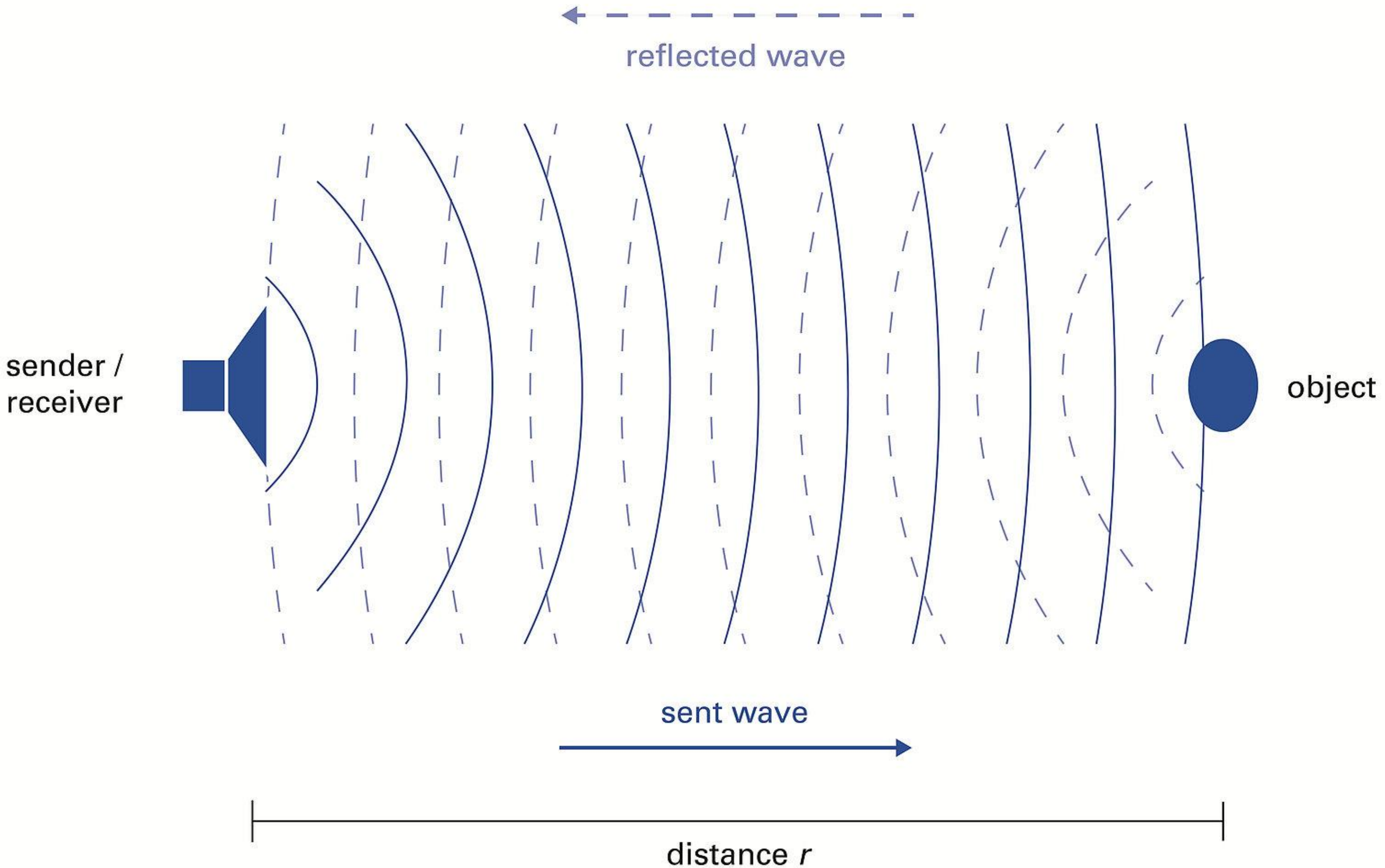
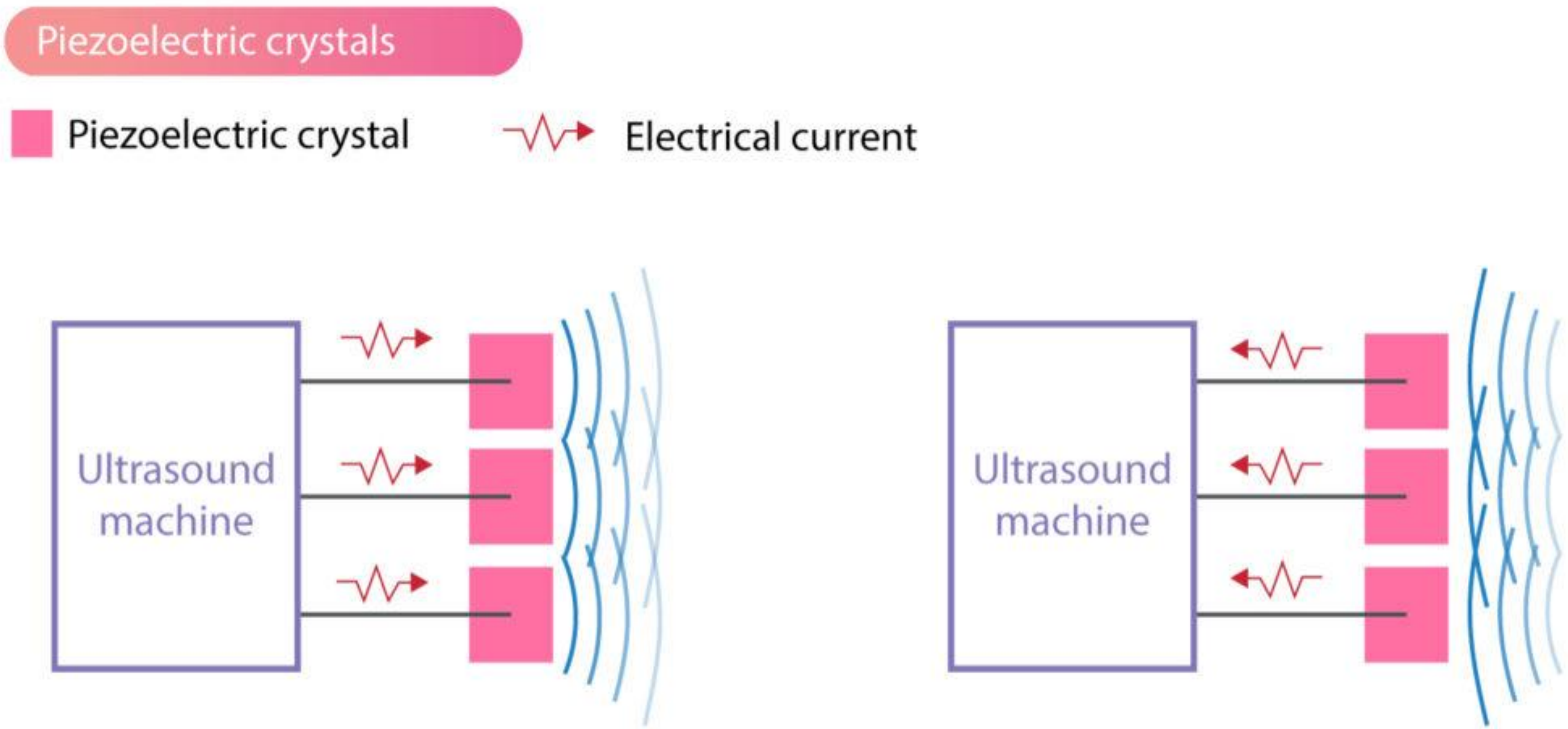
Piezoelectricty



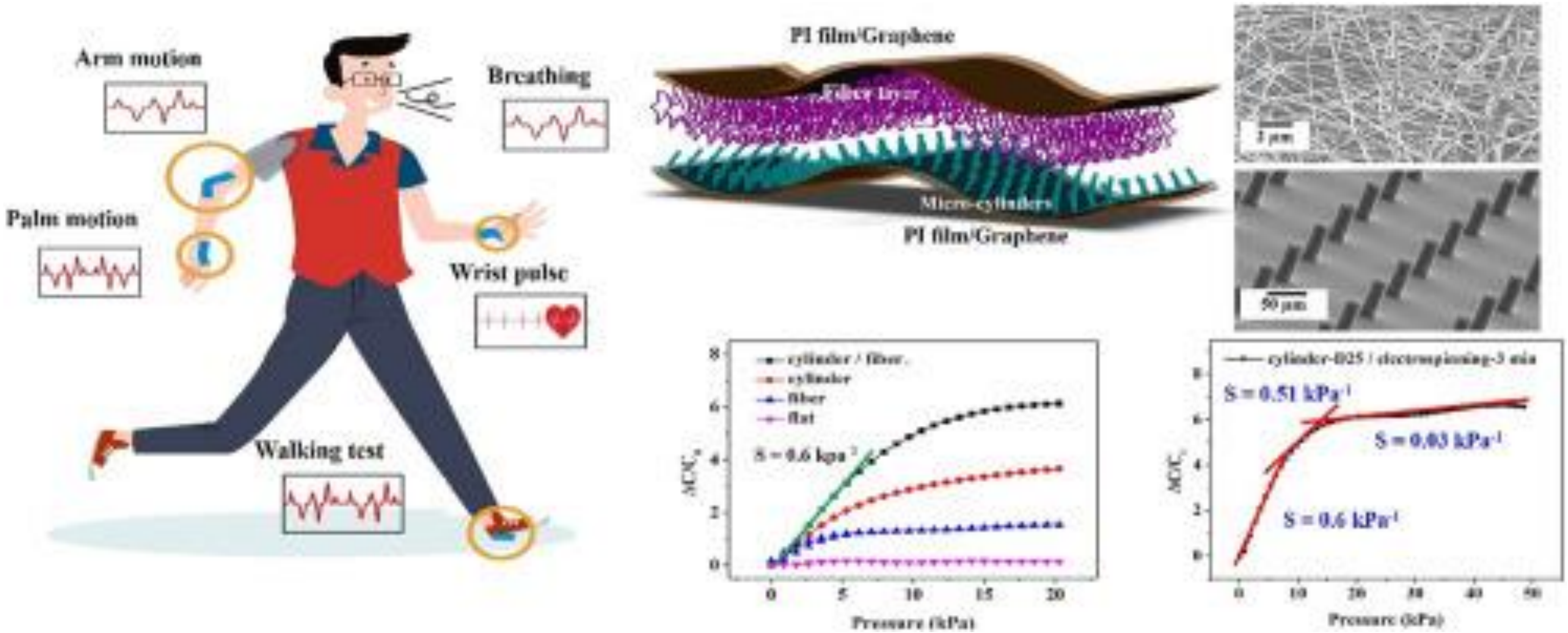
Piezoelectricty



Application of an electrical current to the crystal causes it to vibrate and thus generate ultrasound waves.

Reflected sound waves hit the crystals, causing it to vibrate and generate electrical current that is analyzed by the ultrasound machine.

Piezoelectricty



Piezoelectricty